



Top 14 CAT Miscellaneous LR Questions With Video Solutions

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Questions

Instructions

Mathematicians are assigned a number called Erdos number (named after the famous mathematician, Paul Erdos). Only Paul Erdos himself has an Erdos number of zero. Any mathematician who has written a research paper with Erdos has an Erdos number of 1. For other mathematicians, the calculation of his/her Erdos number is illustrated below:

Suppose that a mathematician X has co-authored papers with several other mathematicians. 'From among them, mathematician Y has the smallest Erdos number. Let the Erdos number of Y be y . Then X has an Erdos number of $y+1$. Hence any mathematician with no co-authorship chain connected to Erdos has an Erdos number of infinity. :

In a seven day long mini-conference organized in memory of Paul Erdos, a close group of eight mathematicians, call them A, B, C, D, E, F, G and H, discussed some research problems. At the beginning of the conference, A was the only participant who had an infinite Erdos number. Nobody had an Erdos number less than that of F.

On the third day of the conference F co-authored a paper jointly with A and C. This reduced the average Erdos number of the group of eight mathematicians to 3. The Erdos numbers of B, D, E, G and H remained unchanged with the writing of this paper. Further, no other co-authorship among any three members would have reduced the average Erdos number of the group of eight to as low as 3.

- At the end of the third day, five members of this group had identical Erdos numbers while the other three had Erdos numbers distinct from each other.
- On the fifth day, E co-authored a paper with F which reduced the group's average Erdos number by 0.5. The Erdos numbers of the remaining six were unchanged with the writing of this paper.
- No other paper was written during the conference.

Question 1

The Erdos number of E at the beginning of the conference was:

- | | |
|----------|---|
| A | 2 |
| B | 5 |
| C | 6 |
| D | 7 |
| E | 8 |

$E \text{ divides } \dots$
 $\text{number} \rightarrow 1$
 $y \rightarrow y+1$
 $\rightarrow 0$

...ven day long mini-conference organized in memory of Paul Erdős, a close group of 10 mathematicians, call them A, B, C, D, E, F, G, and H, discussed some research problems. At the beginning of the conference, A was the only participant who had an infinite Erdős number. Nobody had an Erdős number less than that of F.

On the third day of the conference F co-authored a paper jointly with A and C. On the fourth day, B and D co-authored a paper with A. On the fifth day, the average Erdős number of the group of eight mathematicians to whom A, B, D, E, G and H remained unchanged with the writing of the paper. Further, no mathematician co-authored a paper with more than three members who had an Erdős number of the group of eight to as low as 3.

At the end of the conference, five members of this group had identical Erdős numbers while the other three had Erdős numbers distinct from each other.

On the final day, E co-authored a paper with F which reduced the average Erdős number by 0.5. The Erdős numbers of the remaining seven mathematicians unchanged with the writing of this paper.

No other paper was written during the conference.



Question 2

How many participants had the same Erdos number at the beginning of the conference?

- A 2
- B 3
- C 4
- D 5
- E Cannot be determined

Handwritten notes:

Erdos $\rightarrow 0$
 $x \rightarrow 1$
 $y \rightarrow y+1$
 $\rightarrow D$

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Suppose that a mathematician X has co-authored papers with several other mathematicians. From among them, mathematician Y has the smallest Erdos number. Let the Erdos number of Y be y. Then X has an Erdos number of y+1. Hence any mathematician with no co-authorship chain connected to Erdos has an Erdos number of infinity.

The seven day long mini-conference organized in memory of Paul Erdos, a close group of mathematicians, call them **A, B, C, D, E, F, G and H**, discussed some research problems. At the beginning of the conference, A was the only participant who had an infinite Erdos number. Nobody had an Erdos number less than that of F.

On the third day of the conference F co-authored a paper jointly with A and C. This reduced the average Erdos number of the group of eight mathematicians to 3.5. Further, no other co-authorship among any three members would reduce the average Erdos number of the group of eight to as low as 3.

- At the end of the third day, five members of this group had identical Erdos numbers while the other three had Erdos numbers distinct from each other.
- On the fourth day, E co-authored a paper with F which reduced the group's average Erdos number by 0.5. The Erdos numbers of the remaining members remained unchanged with the writing of this paper.
- No other paper was written during the conference.

VIDEO SOLUTION

Question 3

The Erdos number of C at the end of the conference was:

- A 1
- B 2
- C 3
- D 4
- E 5

Handwritten notes:

Erdos $\rightarrow 0$
 $x \rightarrow 1$
 $y \rightarrow y+1$
 $\rightarrow D$

cracku

Suppose that a mathematician X has co-authored papers with several other mathematicians. From among them, mathematician Y has the smallest Erdos number. Let the Erdos number of Y be y. Then X has an Erdos number of y+1. Hence any mathematician with no co-authorship chain connected to Erdos has an Erdos number of infinity.

The seven day long mini-conference organized in memory of Paul Erdos, a close group of mathematicians, call them **A, B, C, D, E, F, G and H**, discussed some research problems. At the beginning of the conference, A was the only participant who had an infinite Erdos number. Nobody had an Erdos number less than that of F.

On the third day of the conference F co-authored a paper jointly with A and C. This reduced the average Erdos number of the group of eight mathematicians to 3.5. Further, no other co-authorship among any three members would reduce the average Erdos number of the group of eight to as low as 3.

- At the end of the third day, five members of this group had identical Erdos numbers while the other three had Erdos numbers distinct from each other.
- On the fourth day, E co-authored a paper with F which reduced the group's average Erdos number by 0.5. The Erdos numbers of the remaining members remained unchanged with the writing of this paper.
- No other paper was written during the conference.

VIDEO SOLUTION

Question 4

The person having the largest Erdos number at the end of the conference must have had Erdos number (at that time):

- A 5
- B 7
- C 9
- D 14
- E 15

Handwritten notes:

Erdos $\rightarrow 0$

math $\rightarrow 1$

$y \rightarrow y+1$

$\rightarrow 0$

ERDOS NUMBER OF A MATHEMATICIAN IS THE LENGTH OF THE SHORTEST PATH OF CO-AUTHORSHIP FROM HIMSELF TO ERDOS. HIS/HER ERDOS NUMBER IS ILLUSTRATED BELOW:

Suppose that a mathematician X has co-authored papers with several other mathematicians. From among them, mathematician Y has the smallest Erdos number. Let the Erdos number of Y be y. Then X has an Erdos number of y+1. Hence any mathematician with no co-authorship chain connected to Erdos has an Erdos number of infinity.

On the first day of the conference, a group of eight mathematicians, call them A, B, C, D, E, F, G and H, discussed some research problems. At the beginning of the conference, A was the only participant who had an infinite Erdos number. Nobody had an Erdos number less than that of F.

On the third day of the conference F co-authored a paper jointly with A and C. This reduced the average Erdos number of the group of eight mathematicians to 3.5. At the end of the third day, five members of this group had identical Erdos numbers while the other three had Erdos numbers distinct from each other.

On the fourth day, E co-authored a paper with F which reduced the group's average Erdos number by 0.5. The Erdos numbers of the remaining participants remained unchanged with the writing of this paper.

No other paper was written during the conference.

[VIDEO SOLUTION](#)

Question 5

How many participants in the conference did not change their Erdos number during the conference?

- A 2
- B 3
- C 4
- D 5
- E Cannot be determined

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Suppose that a mathematician X has co-authored papers with several other mathematicians. From among them, mathematician Y has the smallest Erdos number. Let the Erdos number of Y be y. Then X has an Erdos number of y+1. Hence any mathematician with no co-authorship chain connected to Erdos has an Erdos number of infinity.

On the third day of the conference F co-authored a paper jointly with A and C. The average Erdos number of the group of eight mathematicians to whom A, B, C, D, E, F, G and H discussed some research problems. At the beginning of the conference, A was the only participant who had an infinite Erdos number. Nobody had an Erdos number less than that of F.

At the end of the third day, five members of this group had identical Erdos numbers while the other three had Erdos numbers distinct from each other.

On the fourth day, E co-authored a paper with F which reduced the group's average Erdos number by 0.5. The Erdos numbers of the remaining members were unchanged with the writing of this paper.

No other paper was written during the conference.

VIDEO SOLUTION

Instructions

Directions for the next 2 questions:

A, B, C are three numbers.

Let $@(A, B)$ = average of A and B,

$/(A, B)$ = product of A and B, and

$X(A, B)$ = the result of dividing A by B

Question 6

Average of A, B and C is given by:

- A $@(/(@/(B, A), 2), C), 3)$
- B $X(@/(@/(B, A), 3), C), 2)$
- C $/(X(@/(B, A), 2), C), 3)$
- D $/(X(@/(@/(B, A), 2), C), 3), 2)$

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Question 7

The sum of A and B is given by:

- A $/(@/(A, B), 2)$
- B $X(@/(A, B), 2)$
- C $@/(A, B), 2)$
- D $@(X(A, B), 2)$

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Instructions

For the following questions answer them individually

Question 8

There were 'x' pigeons and 'y' mynahs in a cage. One fine morning, 'p' birds escaped to freedom. The bird-keeper, knowing only that $p = 7$, was able to figure out without looking into the cage that at least one pigeon had escaped. Which of the following does not represent a possible (x,y) pair?

- A (10,8)
- B (7,2)
- C (25,6)
- D (12,4)

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Question 9

In a hospital there were 200 diabetes, 150 hyperglycaemia and 150 gastro-enteritis patients. Of these, 80 patients were treated for both diabetes and hyperglycaemia. Sixty patients were treated for gastro-enteritis and hyperglycaemia, while 70 were treated for diabetes and gastroenteritis. Some of these patients have all the three diseases. Dr. Dennis treats patients with only gastro-enteritis. Dr. Paul is a generalist. Therefore, he can treat patients with multiple diseases. Patients always prefer a specialist for their disease. If Dr. Dennis had 80 patients, then the other three doctors can be arranged in terms of the number of patients treated as:

- A Paul > Gerard > Hormis
- B Paul > Hormis > Gerard
- C Gerard > Paul > Hormis
- D Cannot be determined

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Question 10

Three labeled boxes containing red and white cricket balls are all mislabeled. It is known that one of the boxes contains only white balls and one only red balls. The third contains a mixture of red and white balls. You are required to correctly label the boxes with the labels red, white and red and white by picking a sample of one ball from only one box. What is the label on the box you should sample?

- A White
- B Red
- C Red and White
- D Not possible to determine from a sample of one ball

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Instructions

A boy is asked to put one mango in a basket when ordered 'One', one orange when ordered 'Two', one apple when ordered 'Three', and is asked to take out from the basket one mango and an orange when ordered 'Four'.

A sequence of orders is given as: 1 2 3 3 2 1 4 2 3 1 4 2 2 3 3 1 4 1 1 3 2 3 4

Question 11

How many total oranges were in the basket at the end of the above sequence?

- A 1
- B 4
- C 3
- D 2

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Question 12

How many total fruits will be in the basket at the end of the above order sequence?

- A 9
- B 8
- C 11
- D 10

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CAT Syllabus PDF



Instructions

Answer the questions based on the following information. The following operations are defined for real numbers.

$a \# b = a + b$, if a and b both are positive else $a \# b = 1$

$a \nabla b = (a \times b)^{a+b}$ if $a \times b$ is positive else $a \nabla b = 1$.

Question 13

$$\frac{((1 \# 1) \# 2) - (10^{1.3} \nabla \log_{10} 0.1)}{1 \nabla 2}$$

- A $\frac{3}{8}$
- B $\frac{4 \cdot \log_{10} 0.1}{8}$
- C $\frac{(4 + 10^{13})}{8}$
- D None of these

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Question 14

$$\frac{(2\#1)}{(1\triangledown2)} =$$

A $\frac{1}{8}$

B 1

C $\frac{3}{8}$

D 3

 [VIEW SOLUTION](#)

